	≈ 24.6 min	
3(a)	If path difference of the sound waves along XY is odd-integer of half-wavelength, the waves meet in antiphase and results in destructive interference and hence minimum intensity.	[1] [1]
3(b)(i)	$BR = \sqrt{12.00^2 + 5.50^2}$ $= 13.20$ m	[1]
3(b)(ii)	R is first order maxima, path difference = 1λ $\lambda = BR - AR$ $= 13.20 - 12.50$ $= 0.70$ m	[1] [1]
3(b)(iii)	$v = f\lambda$ $= 470 \times 0.70$ $= 330 \text{ m s}^{-1}$	[1] [1]
3(c)	Since $I \propto A^2$, for $I' = \frac{I}{3}$, the amplitude of the waves from B is $\frac{1}{\sqrt{3}}A$. At Q, the resultant amplitude of the wave interfering destructively is $A_{\text{resultant}} = A - \frac{1}{\sqrt{3}}A$ $A_{\text{resultant}} \approx 0.423A$ Hence, the resultant intensity is $I_{\text{resultant}} \propto (0.423A)^2 = 0.179I$	[1] amplitude of source B [1] resultant amplitude [1] answer
4(a)(i)	$P = IV$	[1] for